

## Ehmantransport

$$\frac{d}{dz} \left( \nu \frac{dU}{dz} \right) = i f (U - u_0) \quad (1)$$

Integrera:

$$\begin{aligned} \int_0^\infty \frac{d}{dz} \left( \nu \frac{dU}{dz} \right) dz &= \left[ \nu \frac{dU}{dz} \right]_0^\infty = - \left( \nu \frac{dU}{dz} \right)_{z=0} = \\ &= (1) = \int_0^\infty i f (U - u_0) dz \end{aligned} \quad (2)$$

Definiera Ehmantransporten:

$$T_E = \int_0^\infty (U - u_0) dz \quad (3)$$

(2)  $\Rightarrow$

$$T_E = \frac{i}{f} \left( \nu \frac{dU}{dz} \right)_{z=0} \quad (4)$$

Bestäm Ehmantransporten i de båda skikten, från lösningen

$$\begin{cases} U_1 = A_1 e^{(1-i)z/h_1} + B_1 e^{-(1+i)z/h_1} + u_0 & (7) \\ U_2 = B_2 e^{-(1+i)z/h_2} + u_0 & (8) \end{cases}$$

där

$$h_{1,2} = \sqrt{\frac{2\nu_{1,2}}{f}} \quad (9)$$

$$A_1 = -u_0 \frac{1-\varepsilon}{1-\varepsilon + \varepsilon^2(1+\varepsilon)} \quad (10)$$

$$E_1 = -U_0 \frac{E_1^2 (1+\varepsilon)}{1-\varepsilon + E_1^2 (1+\varepsilon)} \quad (11)$$

$$E_2 = -2U_0 \frac{E_1 E_2}{1-\varepsilon + E_1^2 (1+\varepsilon)} \quad (12)$$

oder:

$$\varepsilon = \frac{h_2}{h_1} \quad (13)$$

$$E_{1,2} = e^{(1+i)H/h_{1,2}} \quad (14)$$

$$\left\{ \frac{dU_1}{dz} = \frac{1+i}{h_1} \left( A_1 e^{(1+i)z/h_1} - B_1 e^{-(1+i)z/h_1} \right) \right. \quad (15)$$

$$\left. \frac{dU_2}{dz} = -\frac{1+i}{h_2} B_2 e^{-(1+i)z/h_2} \right. \quad (16)$$

Vi ben sedan tillgärde

$$\left. \frac{dU_1}{dz} \right|_{z=0} = U_0 \frac{1+i}{h_1} \frac{E_1^2 - R}{E_1^2 + R} = U_0 \frac{1+i}{h_1} \frac{E_1^2 (1+\varepsilon) - (1-\varepsilon)}{1-\varepsilon + E_1^2 (1+\varepsilon)} \quad (17)$$

där

$$R = \frac{1-\varepsilon}{1+\varepsilon} \quad (18)$$

$$(15) \Rightarrow$$

$$\left. \frac{dU_1}{dz} \right|_{z=H} = \frac{1+i}{h_1} \left( A_1 E_1 - \frac{B_1}{E_1} \right) = [(10), (11)] =$$

$$= \frac{1+i}{h_1} \left( -U_0 \frac{E_1 (1-\varepsilon) - E_1 (1+\varepsilon)}{1-\varepsilon + E_1^2 (1+\varepsilon)} \right) =$$

$$= \frac{1+i}{h_1} U_0 \frac{2\varepsilon E_1}{1-\varepsilon + E_1^2 (1+\varepsilon)} \quad (19)$$

(16)  $\Rightarrow$

$$\left. \frac{dU_2}{dz} \right|_{z=H} = - \frac{1+i}{h_2} \frac{B_2}{E_2} = [(12)] =$$

$$= - \frac{1+i}{h_2} \frac{2U_0 E_1}{1-\varepsilon + E_1^2(1+\varepsilon)} \quad (20)$$

Satt

$$T_1 = \int_0^H (u - u_0) dz$$

$$T_2 = \int_H^\infty (u - u_0) dz$$

(1)  $\Rightarrow$

$$T_1 = -i \frac{\nu}{f} \left[ \frac{dU_1}{dz} \right]_0^H = -i \frac{h_1^2}{2} \left[ \frac{dU_1}{dz} \right]_0^H \quad (21)$$

$$T_2 = -i \frac{\nu}{f} \left[ \frac{dU_2}{dz} \right]_H^\infty = i \frac{h_2^2}{2} \left. \frac{dU_2}{dz} \right|_H \quad (22)$$

(17) och (19) i (21)  $\Rightarrow$

$$\begin{aligned} T_1 &= -i \frac{h_1^2}{2} U_0 \frac{1+i}{h_1} \frac{2\varepsilon E_1 - E_1^2(1+\varepsilon) + 1 - \varepsilon}{1 - \varepsilon + E_1^2(1+\varepsilon)} = \\ &= - \frac{1-i}{2} U_0 h_1 \frac{\varepsilon(-1 + 2E_1 - E_1^2) + 1 - E_1^2}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} = \\ &= - \frac{1-i}{2} U_0 h_1 \frac{\varepsilon(E_1 - 1)^2 + E_1^2 - 1}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} = \\ &= - \frac{1-i}{2} U_0 h_1 (E_1 - 1) \frac{\varepsilon(E_1 - 1) + E_1 + 1}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} \quad (23) \end{aligned}$$

(20) i (22)  $\Rightarrow$

$$T_2 = i \frac{h_2^2}{2} \frac{1+i}{h_2} \frac{2\mu_0 E_1}{1-\varepsilon + E_1^2(1+\varepsilon)} =$$

$$= - \frac{1-i}{2} \mu_0 h_2 \frac{2E_1}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} \quad (24)$$

Totala Elemen transporten:

$$T_E = T_1 + T_2 = - \frac{1-i}{2} \mu_0 \frac{h_1 [ \varepsilon(E_1 - 1)^2 + E_1^2 - 1 ] + 2h_2 E_1}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} =$$

$$= - \frac{1-i}{2} \mu_0 h_1 \frac{\varepsilon(E_1^2 - 2E_1 + 1) + E_1^2 - 1 + 2\varepsilon E_1}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} \Rightarrow$$

$$T_E = - \frac{1-i}{2} \mu_0 h_1 \frac{E_1^2 - 1 + \varepsilon(E_1^2 + 1)}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} \quad (25)$$

Samma resultat som för direkt från (4):

$$T_E = i \frac{h_1^2}{2} \frac{dU_1}{dz} \Big|_{z=0} = \{ (17) \} =$$

$$= i \frac{h_1^2}{2} \mu_0 \frac{1+i}{h_1} \frac{E_1^2(1+\varepsilon) - 1 + \varepsilon}{1-\varepsilon + E_1^2(1+\varepsilon)} =$$

$$= - \frac{1-i}{2} \mu_0 h_1 \frac{E_1^2 - 1 + \varepsilon(E_1^2 + 1)}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)}$$

sätt in

$$\varphi = \frac{\mu}{h_1} \quad (26)$$

och anta att  $\varphi \ll 1$  och  $\varepsilon \ll 1$ . Vi får

$$E_1 = e^{(1+i)\varphi} \approx 1 + (1+i)\varphi + i\varphi^2 \dots \quad (27)$$

(23) och (24) ger då

$$\begin{aligned}
 T_1 &\approx -\frac{1-i}{2} u_0 h_1 (1+i) \varphi \frac{\varepsilon(1+i)\varphi + 2 + (1+i)\varphi}{2 + 2(1+i)\varphi + \varepsilon 2(1+i)\varphi} \\
 &\approx -\frac{1}{2} u_0 h_1 \varphi \frac{2 + (1+i)(1+\varepsilon)\varphi}{1 + (1+i)(1+\varepsilon)\varphi} \approx \\
 &\approx -u_0 h_1 \varphi \frac{1 + \frac{1}{2}(1+i)(1+\varepsilon)\varphi}{1 + (1+i)(1+\varepsilon)\varphi} \approx \\
 &\approx -u_0 h_1 \varphi \left[ 1 - \frac{1}{2}(1+i)(1+\varepsilon)\varphi \right] = \quad (28)
 \end{aligned}$$

$$\approx -u_0 H \left[ 1 - \frac{1}{2}(1+i)(1+\varepsilon) \frac{H}{h_1} \right] \quad (29)$$

$$\begin{aligned}
 T_2 &\approx -\frac{1-i}{2} u_0 h_2 \frac{2[1 + (1+i)\varphi]}{2 + 2(1+i)\varphi + \varepsilon 2(1+i)\varphi} \approx \\
 &\approx -\frac{1-i}{2} u_0 h_2 \frac{1 + (1+i)\varphi}{1 + (1+i)(1+\varepsilon)\varphi} \approx \\
 &\approx -\frac{1-i}{2} u_0 h_2 [1 + (1+i)\varphi - (1+i)(1+\varepsilon)\varphi] = \\
 &\approx -\frac{1-i}{2} u_0 h_2 \varepsilon [1 - (1+i)\varepsilon\varphi] \quad (30)
 \end{aligned}$$

(25)  $\Rightarrow$

$$\begin{aligned}
 T_0 &\approx -\frac{1-i}{2} u_0 h_1 \frac{2(1+i)\varphi + \varepsilon(2 + 2(1+i)\varphi)}{2 + 2(1+i)\varphi + \varepsilon 2(1+i)\varphi} = \\
 &= -\frac{1-i}{2} u_0 h_1 \frac{\varepsilon + (1+i)\varphi + \varepsilon(1+i)\varphi}{1 + (1+i)\varphi + (1+i)\varepsilon\varphi} \approx \\
 &\approx -\frac{1-i}{2} u_0 h_1 \varepsilon \frac{1 + (1+i)\frac{\varphi}{\varepsilon} + (1+i)\varphi}{1 + (1+i)\varphi + (1+i)\varepsilon\varphi} \approx \\
 &\approx -\frac{1-i}{2} u_0 h_1 \varepsilon \left( 1 + (1+i)\frac{\varphi}{\varepsilon} + (1+i)\varphi - (1+i)\varphi - (1+i)\varepsilon\varphi \right) \\
 &\approx -\frac{1-i}{2} u_0 h_1 \left[ \varepsilon + (1+i)H \right] = \\
 &\approx -\frac{1-i}{2} u_0 \left[ h_2 + (1+i)H \right] \quad (31)
 \end{aligned}$$

Notera (28) och (30):

$$\begin{aligned} T_1 + T_2 &\approx -u_0 h_1 \varphi \left[ 1 - \frac{1}{2}(1+i)(1+\varepsilon)\varphi \right] - \\ &\quad - \frac{1-i}{2} u_0 h_2 \varepsilon \left[ 1 - (1+i)\varepsilon\varphi \right] = \\ &= -\frac{1-i}{2} u_0 h_2 - u_0 h_1 \left[ \varphi - \frac{1}{2}(1+i)(1+\varepsilon)\varphi^2 - \varepsilon^2\varphi \right] = \\ &= -\frac{1-i}{2} u_0 h_2 - u_0 H + u_0 h_1 \left[ \frac{1}{2}(1+i)(1+\varepsilon)\varphi^2 + \varepsilon^2\varphi \right] \end{aligned} \quad (32)$$

De första två termerna ger här (31).

Notera också att för  $\varepsilon=1 \Rightarrow \varepsilon_1 = \varepsilon_2 = \varepsilon$  så  
från (25):

$$T_E = -\frac{1-i}{2} u_0 h_E$$

Detta stämmer med första termen i (32)  
och (30),